

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2025

Bachelor in Computer Applications

Course Title: **OOP in Java**

Code No: **CACS204** Semester: **III**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. What is command-line arguments? Write a program to accept two numbers and display their sum using command-line arguments.
3. Write a program to demonstrate exception handling of `ArrayIndexOutOfBoundsException`.
4. What is string immutability? Explain with an example.
5. Write a program to create multiple threads and execute them concurrently.
6. Explain the `Serializable` interface in java with an example.
7. What is multiple inheritance? Explain how multiple inheritance is implemented in Java?
8. What is JDBC? Explain its architecture.

Group C

Attempt any TWO questions[2x10=20]

9. Explain nested and inner classes in java with examples. Discuss the scenarios where inner classes are useful.
10. Create a class `Person` (with attributes `name` and `age`, and a method `display()`). Derive a class `Employee` from `Person` with additional attributes `employeeId` and `salary`. Use constructors and override the `display()` method to include all details.
11. Build a form with `Passenger Name`, `Flight Number`, `Seat Preference` (`Window`, `Aisle`, `Middle`), and a checkbox for "I agree to baggage rules." Clicking `Confirm Booking` should save data to "flight.txt" if rules are accepted otherwise display "Accept baggage rules first."